Bruce Xi

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brucexi999

Software engineer focused on ML training systems, performance optimization, and evaluation reliability. Hands-on experience with LLM post-training, accelerator kernels, and debugging across numerical and compiler layers.

Experience

Huawei Canada **Markham, ON, Canada**

SW/HW Co-Design Engineer *Jan. 2025 – Present*

- Build and optimize ML training and inference software on Ascend NPUs, spanning LLM post-training workflows, PyTorch integration, and custom Ascend C/Triton kernels.
- Use profiling, benchmarking, roofline analysis, and numerical validation to diagnose performance and correctness issues; contribute to LLVM/MLIR compiler integration and software/hardware co-design for GDN and Mamba models.
- Bring up Qwen and DeepSeek models on MindSpeed-LLM; tune micro-batch sizes and 3D parallelism to address memory and throughput bottlenecks.

Projects

Post-Training for a Triton Kernel Agent

Huawei Canada *Completed Oct. 2026*

Collaborated on supervised fine-tuning (SFT) and reinforcement learning to train an LLM-based agent that translates PyTorch reference operators into correct, efficient Triton kernels on Ascend NPUs through iterative coding, verification, and optimization.

- Distilled and curated multi-turn kernel-development trajectories for **SFT**, filtering examples by correctness and measured speedup; removed unproductive iterations and compressed reasoning to fit the training token budget.
- Compared agent harnesses and hardware-specific guidance through controlled ablations to inform the RL training setup. On 18 KernelBench matrix-multiplication tasks, guidance increased Pass@1 from **13/18 to 17/18** and Fast@1 from **0/18 to 9/18** against eager PyTorch.
- Added **profiler feedback** and optimization guidance to the agent's toolset for RL training, enabling it to diagnose hardware bottlenecks and refine kernels. One matrix-multiplication case achieved **24% lower latency** (2,886 to 2,197 μs).
- Improved the validity of execution feedback by tracing convolution failures to the reference's HF32 compute mode versus full FP32. Inspected training trajectories to identify four prompt/tool-interface issues and propose fixes.
- **Accelerated the agent's RL training by 25%** with partial rollout in verl/vLLM, reducing straggler waiting through cross-batch scheduling, resumable trajectories, and importance-sampling correction across policy updates.

Project outputs: ATRL (ICLR 2027 submission), a Huawei technical report, and a contribution to verl-ascend-recipe.

Projects (continued)

Triton-to-NPU Compiler Debugging (LLVM/MLIR)

Huawei Canada *Feb. 2026 – Apr. 2026*

- **Resolved silent data corruption** in Triton-to-NPU compilation by tracing failures through **600+ compiler pass dumps** to incorrect hardware-predicate materialization in MLIR-to-LLVM lowering.
- Used intermediate-value logging and SSA manipulation to isolate corrupted vector lanes; implemented the fix and regression tests in AscendNPU-IR.

Training Performance Optimization for a Mamba2 Model

Huawei Canada *Mar. 2025 – Dec. 2025*

- Developed Ascend C kernels and integrated them into PyTorch via Pybind. Fused memory-bound elementwise operators to increase arithmetic intensity; a kernel achieved **12× forward and 15× backward speedups**, reducing overall training-step time by **4.6%** and improving MFU by **1.5%**.
- Optimized workload partitioning, tiling, L2 cache reuse, and memory-access patterns; used ping-pong buffering to overlap computation and data transfers within on-chip buffer limits.
- Validated numerical correctness and analyzed performance using Ascend's debugger, profiler, and simulator; diagnosed out-of-memory errors, synchronization faults, and pipeline bubbles.

Education

The University of British Columbia **Vancouver, BC, Canada**

Master of Engineering in Electrical and Computer Engineering, GPA: 4.0 *2021 – 2024*

Relevant courses: Computer Architecture, VLSI, ML Accelerator, Deep Learning

Imperial College London **London, UK**

Bachelor of Engineering in Materials Science and Engineering, First-Class Honours *2018 – 2021*

Dean's List for three consecutive years (2018–2021)

Skills

Languages & frameworks: C/C++, Python, PyTorch, Triton, Ascend C

Training systems: LLM post-training, SFT data generation and curation, trajectory distillation, RL rollout scheduling; verl, vLLM, GRPO

Evaluation & tooling: Agent tools and harness integration, controlled experiments and ablations, numerical validation, trajectory failure analysis

Performance engineering: Profiling and benchmarking, roofline analysis, numerical correctness, kernel fusion, tiling, compute/memory overlap

Compiler integration: LLVM/MLIR, lowering pipelines, IR-level debugging, regression testing

Models & systems: GDN and Mamba architectures, Ascend NPUs, software/hardware co-design, distributed training and parallelism configuration with MindSpeed-LLM